

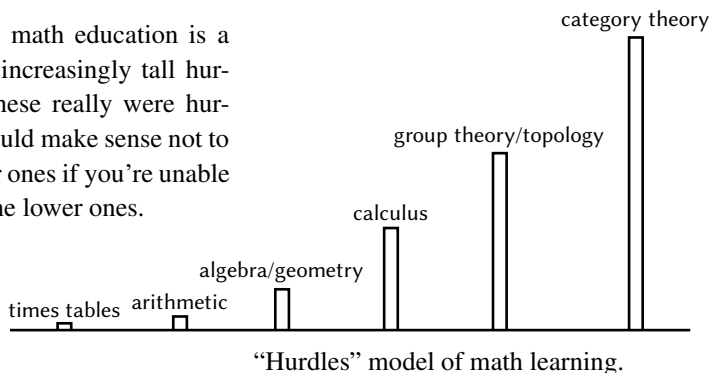
attention on what is relevant for a particular point of view and temporarily disregards the rest so that we can get to the heart of a structure or an argument. In making these connections and finding these deep structures we package up intractably complex situations into succinct units, enabling us to address yet more complicated situations and use our limited brain power to greater effect. This starts with numbers, where instead of saying “ $1 + 1$ ” all the time we can call it 2, or we fit squares together and call the result a cube, and then build up to more complex mathematical structures as we’ll see throughout this book.

This is what I think the power and importance of abstract mathematics are. The idea that it is relevant to the whole of life and thus illuminating for everyone may be surprising, but is demonstrated by the wide range of examples that I have found where category theory helps, despite the field being considered perhaps the “most abstract” of all mathematics. This includes examples such as privilege, sexism, racism, sexual harassment. These are not the sort of contrived real life examples involving the purchase of 17 watermelons, but are *real* real life questions, things we actually do (or should) think about in our daily lives.

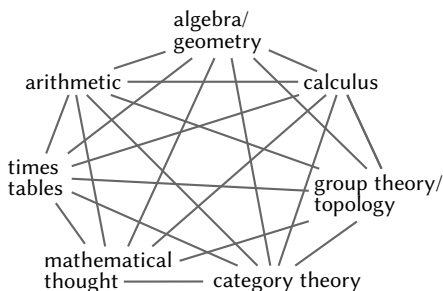
If people are put off math then they are put off these ways of thinking that could really intrigue and help them. The sad part is that they are put off an entirely different kind of math usually involving algorithms, formulae, memorization and rigid rules, which is not what this abstract math is about at all. Math is misunderstood, and the first impression many people get of it is enough to put them off, forever, something that they might have been able to appreciate and benefit from if they saw it in its true light.

Traditional mathematics: subjects

A typical math education is a series of increasingly tall hurdles. If these really were hurdles it would make sense not to try higher ones if you’re unable to clear the lower ones.



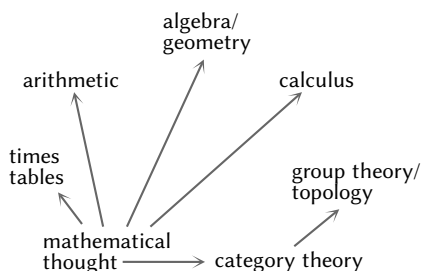
However, math is really more like an interconnected web of ideas, perhaps like this; everything is connected to everything else, and thus there are many possible routes around this web depending on what sort of brain you have.



“Interconnectedness” in math learning.

Some people do need to build up gradually through concrete examples towards abstract ideas. But not everyone is like that. For some people, the concrete examples don’t make sense until they’ve grasped the abstract ideas or, worse, the concrete examples are so offputting that they will give up if presented with those first. When I was first introduced to single malt whisky I thought I didn’t like it, but I later discovered it was because people were trying to introduce me “gently” via single malts they considered “good for beginners”. It turns out I only like the extremely smoky single malts of Islay, not the sweeter, richer ones you might be expected to acclimatize with.

I am somewhat like that with math as well. My route through the web of mathematics was something like this diagram.



My progress to higher level mathematics did not use my knowledge of mathematical subjects I was taught earlier. In fact after learning category theory I went back and understood everything again and much better.

I have confirmed from several years of teaching abstract mathematics to art students that I am not the only one who prefers to use abstract ideas to illuminate concrete examples rather than the other way round. Many of these art students consider that they’re bad at math because they were bad at memorizing times tables, because they’re bad at mental arithmetic, and they can’t solve equations. But this doesn’t mean they’re bad at math — it just means they’re not very good at times tables, mental arithmetic and equations, an absolutely tiny part of mathematics that hardly counts as abstract at all. It turns out that they do not struggle nearly as much when we get to abstract things such as